Extending nn-UNet for Brain Tumor Segmentation

Intelligent Analysis of Biomedical Images

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Introduction

Introduction & Importance of the Problem

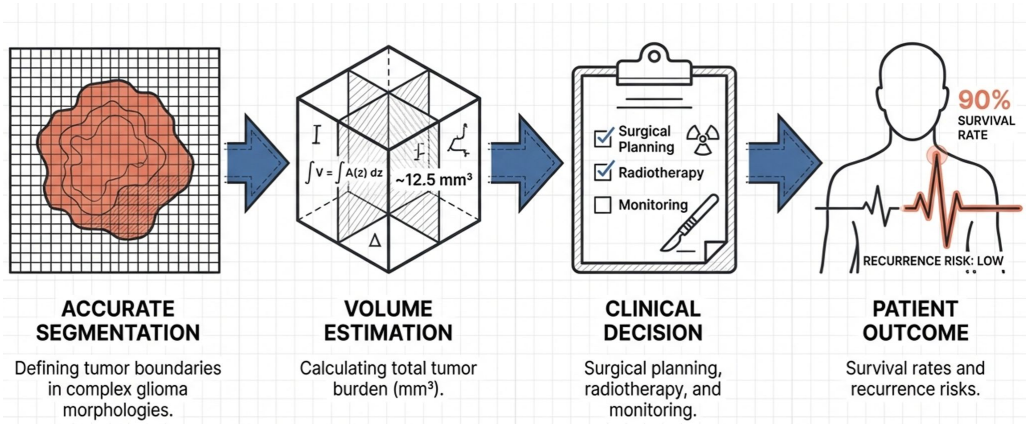
Detecting and analyzing brain tumors in MRI is a **critical task** in medical imaging.

Accurate localization of the tumor directly affects **diagnosis**, **treatment planning**, and **patient outcomes**.

Gliomas have a **heterogeneous appearance** and **irregular shapes** → making **automatic segmentation** difficult.

Precise segmentation helps with:

- **Estimating tumor size and volume**
- **Treatment planning / radiotherapy planning**
- **Monitoring disease progression**



Limitations of Previous Methods & Research Gap

Many previous studies mainly focused on **feature extraction** and **classification**.

These methods often do not produce an accurate **spatial map (segmentation mask)** → limiting **clinical usability** and **interpretability**.

Without reliable segmentation:

- **Volume measurement is not dependable**
- **Tracking tumor growth over time becomes harder**
- There is no precise **map** for **surgery** or **radiotherapy planning**

Clinical Motivation: Tumor Size

Tumor size at the time of diagnosis is associated with important clinical outcomes, such as:

- **Survival rate**
- **Risk of recurrence**
- **Neurological deficits**
- **Length of hospitalization**

Proposed Work Idea

Objective: Improve an existing baseline by adding a **lightweight, threshold-based segmentation** stage.

Why this approach?

- Improve **tumor localization** and **interpretability** of the segmented region
- Maintain **low computational cost** (lightweight and practical)

Positioning of the work: A bridge between

- **Classical intensity-based methods** (simple and interpretable, but limited)
- **Deep learning methods** (powerful, but sometimes less interpretable and computationally heavy)

Related Works

Classical Brain Tumor Segmentation

- Early methods used intensity-based + morphological operations:
  - Thresholding, region growing, clustering (K-means, Fuzzy C-means).

Advantages: fast, computationally efficient, and interpretable.

- Limitations: struggle with
  - heterogeneous tumors, low-contrast regions, intensity inhomogeneity,
  - high sensitivity to manual parameter tuning.

- Robustness improvements introduced:
  - anatomical priors and symmetry-based analysis → better structural awareness,
  - but still noise-sensitive and fail when tumor shapes deviate from expected patterns.

Classical Brain Tumor Segmentation

Statistical modeling example:

- Pravitasari et al.: GMM + EM for segmentation + 3D reconstruction → enables volume estimation.
- Limitation: intensity-only modeling degrades with irregular boundaries and heterogeneous textures.
- Our contrast: nnU-Net learns spatial + contextual features → more stable boundaries and volume.

Hybrid classical pipeline:

- Basthikodi et al.: improved K-means + FCM + SVM + ANN (multi-stage detection/localization/staging).
- Limitation: relies on handcrafted features → weaker scalability/generalization.
- Our contrast: fully automated deep framework → reduced parameter sensitivity.

Deep Learning–Based Methods

CNNs significantly improved medical image segmentation.

U-Net and variants became the standard in biomedical segmentation.

nnU-Net advances this by self-configuring:

- adapts preprocessing, architecture, and training to the dataset,
- achieves strong performance on benchmarks like BraTS.

Remaining challenges:

- accurate segmentation of small tumors and low-contrast areas.

Data-scarce setting example:

- Zhang et al.: supervised/transfer/self-supervised learning for small metastatic tumors (33 MRI cases),
- used U-Net and Swin UNETR,
- Dice for small tumors still relatively low → fine boundaries remain difficult.

Deep Learning–Based Methods

- Our contrast: improve refinement within nnU-Net by adding a lightweight threshold-based refinement stage
  - boosts spatial precision without heavy training strategies.
  -
- Clinical evaluation example:
  - Barhoumi et al.: AI segmentation for glioblastoma (T1c & FLAIR) comparable to expert neuroradiologists,
  - showed low inter-user variability in volumetric measurements.
  - Our contrast: focuses on improving an open, self-configuring framework (nnU-Net) + interpretable refinement.

Tumor Volume Estimation & Clinical Significance

Precise tumor volume matters for:

- **treatment planning**, **monitoring therapy response**, and **prognosis**.

Manual volume estimation is often approximate:

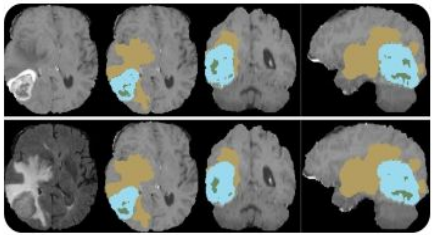
- uses geometric assumptions → **inter-observer variability** and lower reproducibility.
- Classical/statistical methods (e.g., **GMM-based ROI**) enabled early 3D reconstruction and volume computation.
- Modern AI improves agreement vs. manual contouring, but:
- accurate volume **depends directly on accurate boundaries**,
- especially with **irregular shapes**, **small lesions**, and **low-contrast MRI**.

Tumor Volume Estimation & Clinical Significance

- Our approach: combine
  - nnU-Net’s robust adaptive segmentation + a threshold-based refinement step
  - bridging classical intensity-based concepts with deep representations
  - improves interpretability while keeping automation and efficiency.
- Clinical relevance:
  - tumor size at detection correlates with survival, recurrence risk, neurological deficits, and hospitalization duration.

BRaTS 2021 Task 1 Dataset

RSNA-ASNR-MICCAI Brain Tumor Segmentation (BraTS) Challenge 2021



Data Card | Code (109) | Discussion (5) | Suggestions (0)

About Dataset

Description

Ample multi-institutional routine clinically-acquired multi-parametric MRI (mpMRI) scans of glioma, with pathologically confirmed diagnosis and available MGMT promoter methylation status, are used as the training, validation, and testing data for this year's BraTS challenge.

Specifically, for Task 1 the datasets used in this year's challenge have been updated, since BraTS'20, with many more routine clinically-acquired mpMRI scans. Ground truth annotations of the tumor sub-regions are created and approved by expert neuroradiologists for every subject included in the training, validation, and testing datasets to quantitatively evaluate the predicted tumor segmentations.

Imaging Data Description

Usability

8.75

License

Data files © Original Authors

Expected update frequency

Never

Tags

Business

BraTS: Brain Tumor Segmentation Challenge

Annual international benchmark for brain tumor segmentation, established in **2012**

Focuses on **glioma segmentation** from **multi-parametric MRI**

Provides **large-scale, multi-institutional, fully annotated datasets**

Data (BraTS 2021)

Dataset: Multi-parametric MRI scans from **2000 glioma patients**

1251 cases with labels for training

219 cases for public validation leaderboard

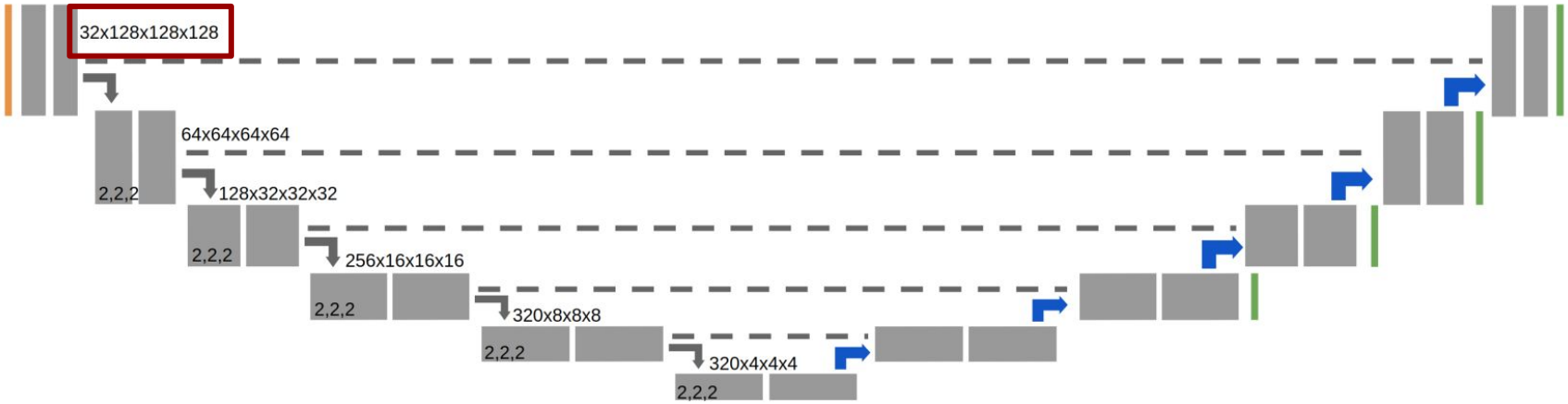
530 cases for private test leaderboard

Annotations:

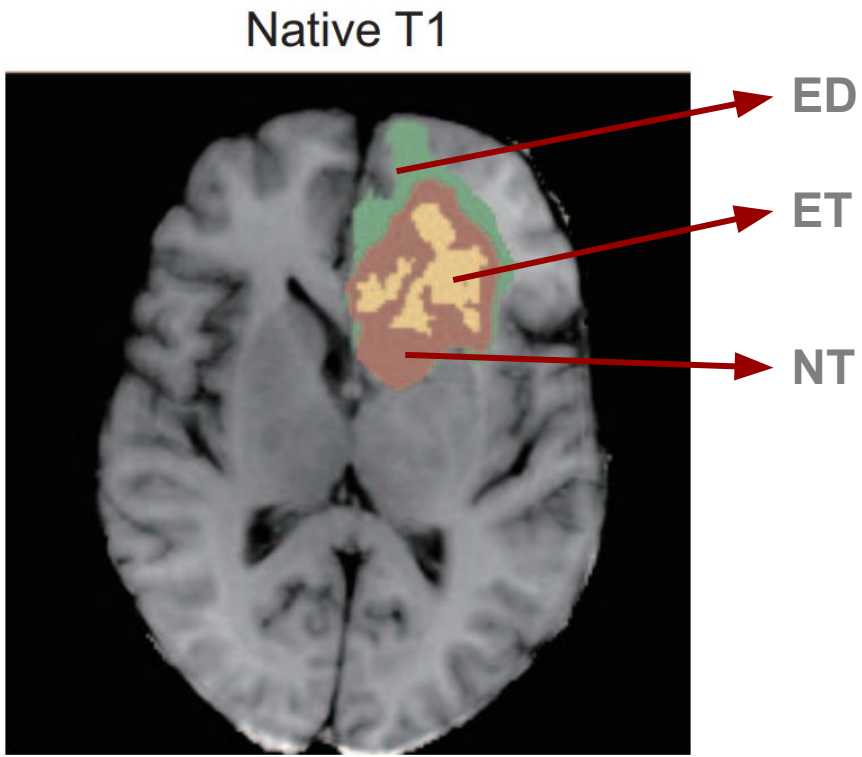
Final approval by **expert neuro-radiologists**

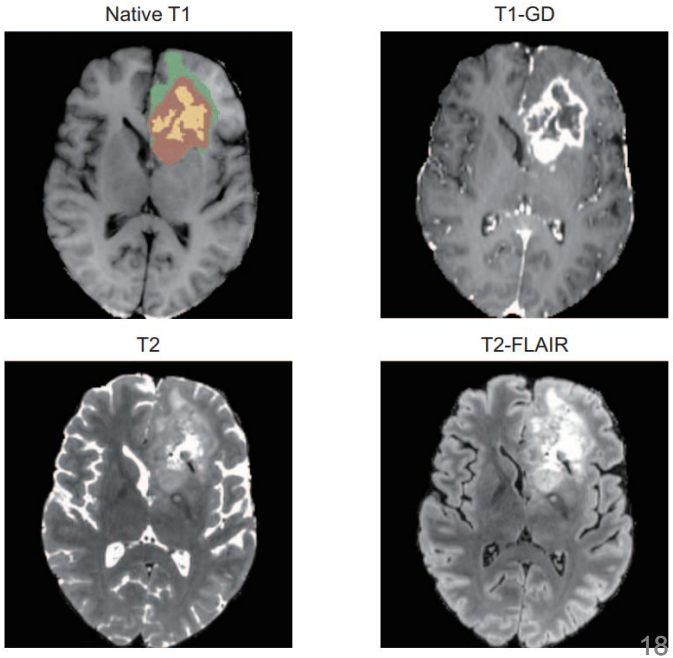
Base model: nnU-Net

- Winning entry for **BraTS 2020**
- 3D U-Net Core
- Leaky ReLU + **batch norm** after every conv
- mpMRI volumes used as 4-channels input
- **Deep supervision** by outputs at most of resolutions

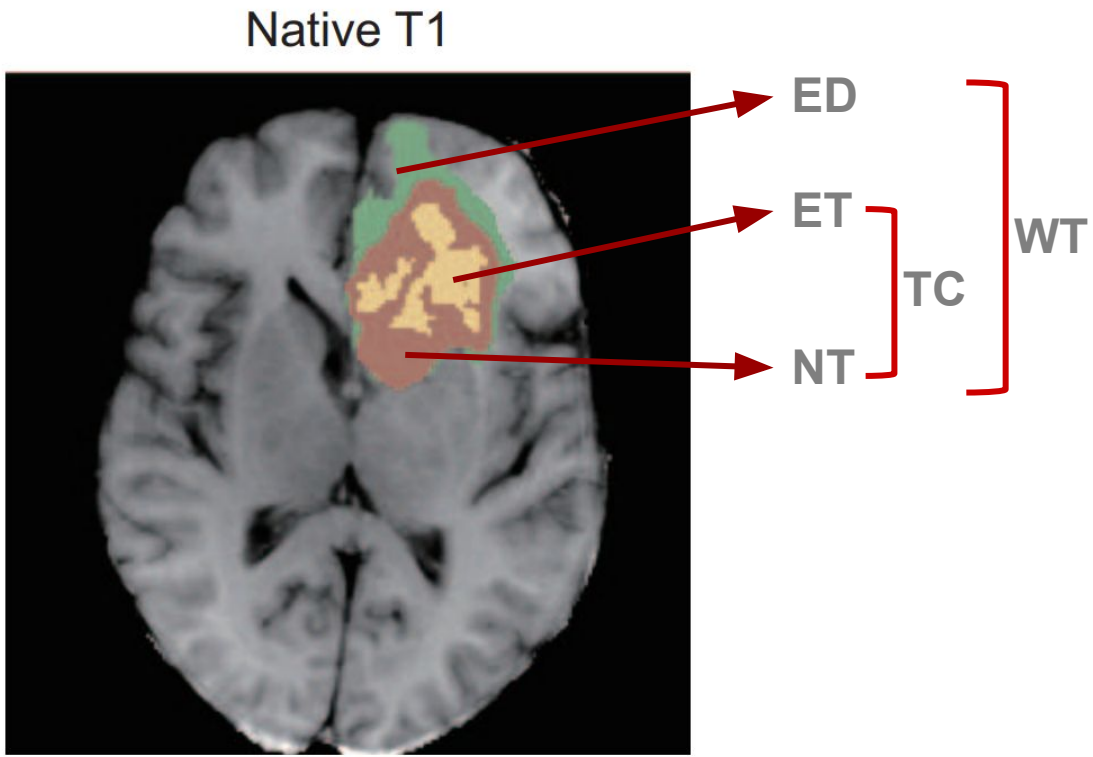


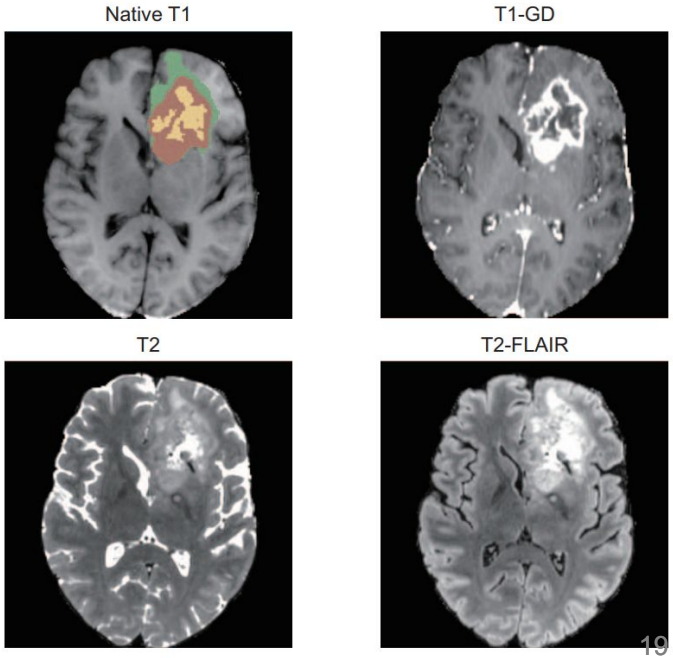
nnU-Net, region-based training



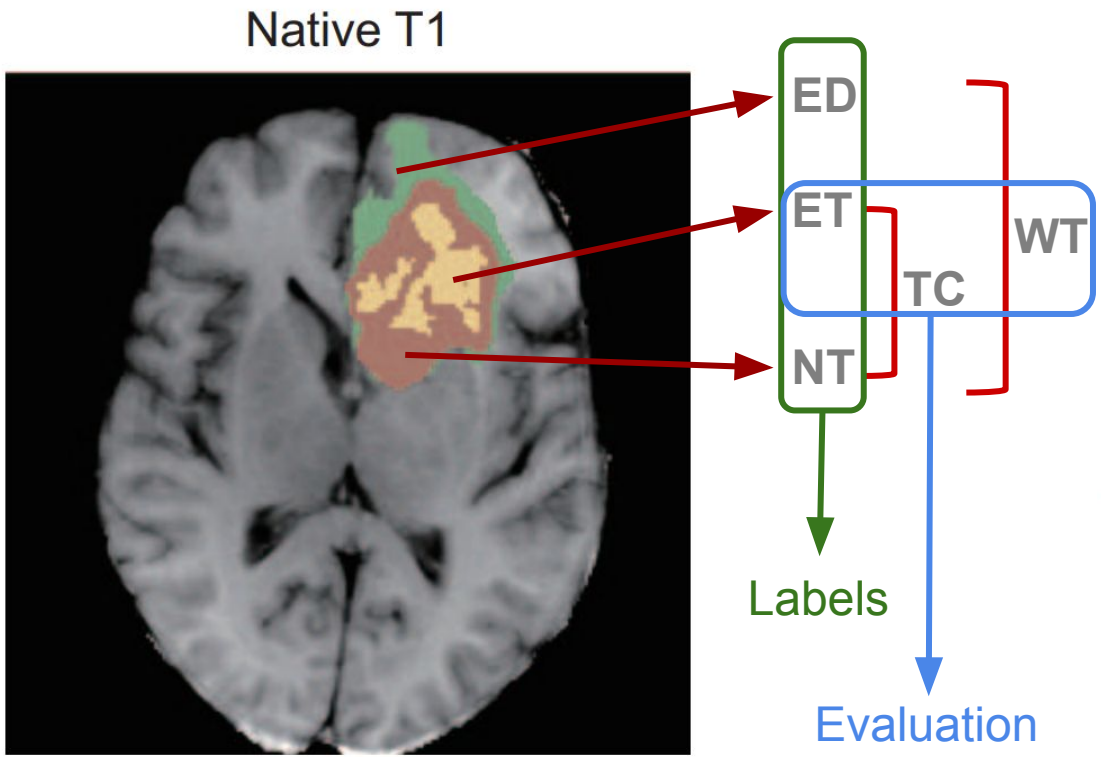


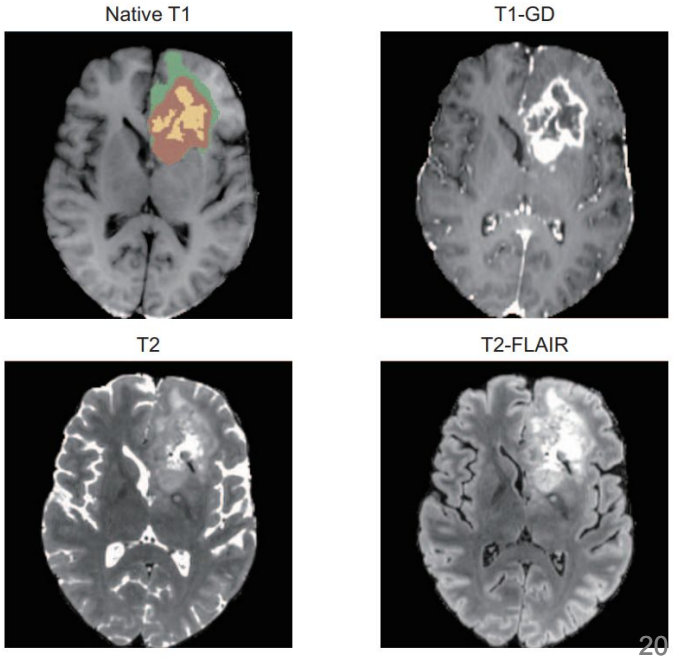
nnU-Net, region-based training





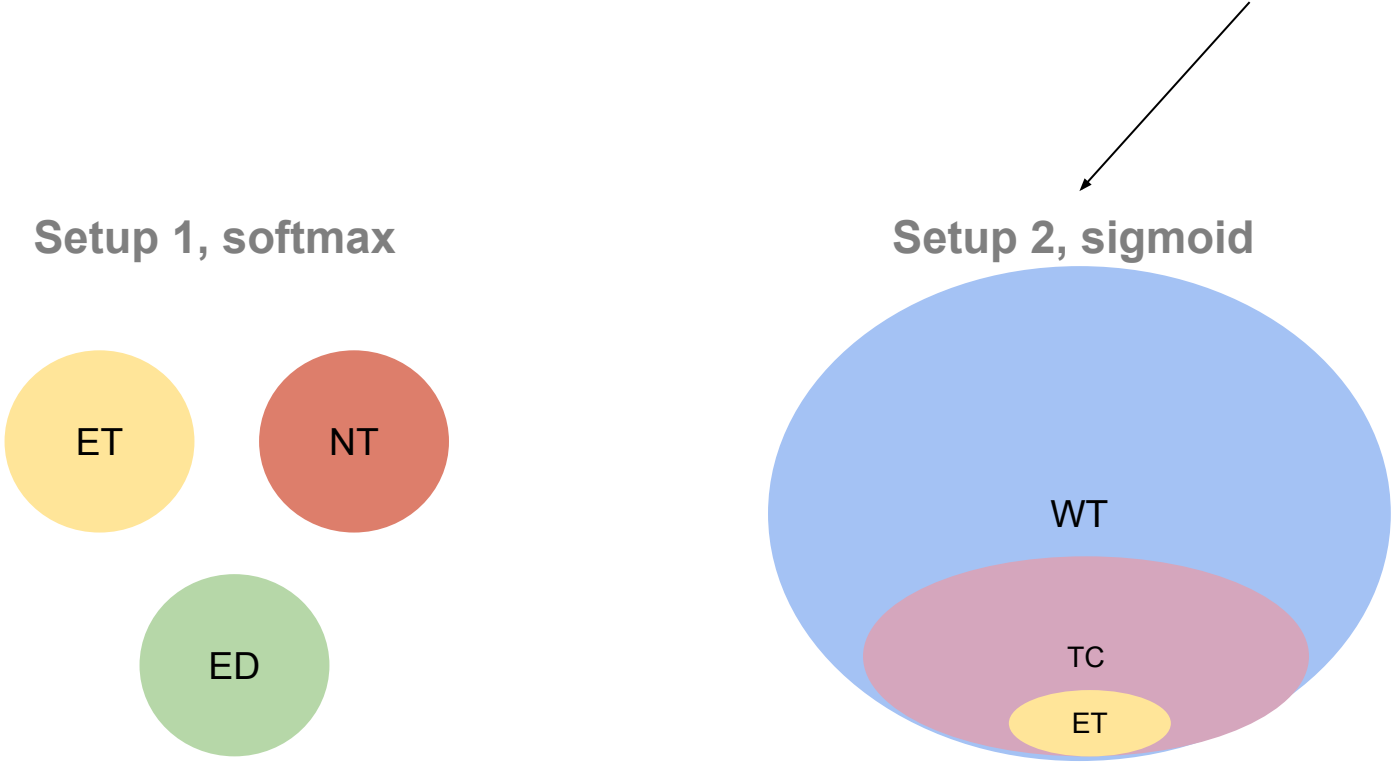
nnU-Net, region-based training





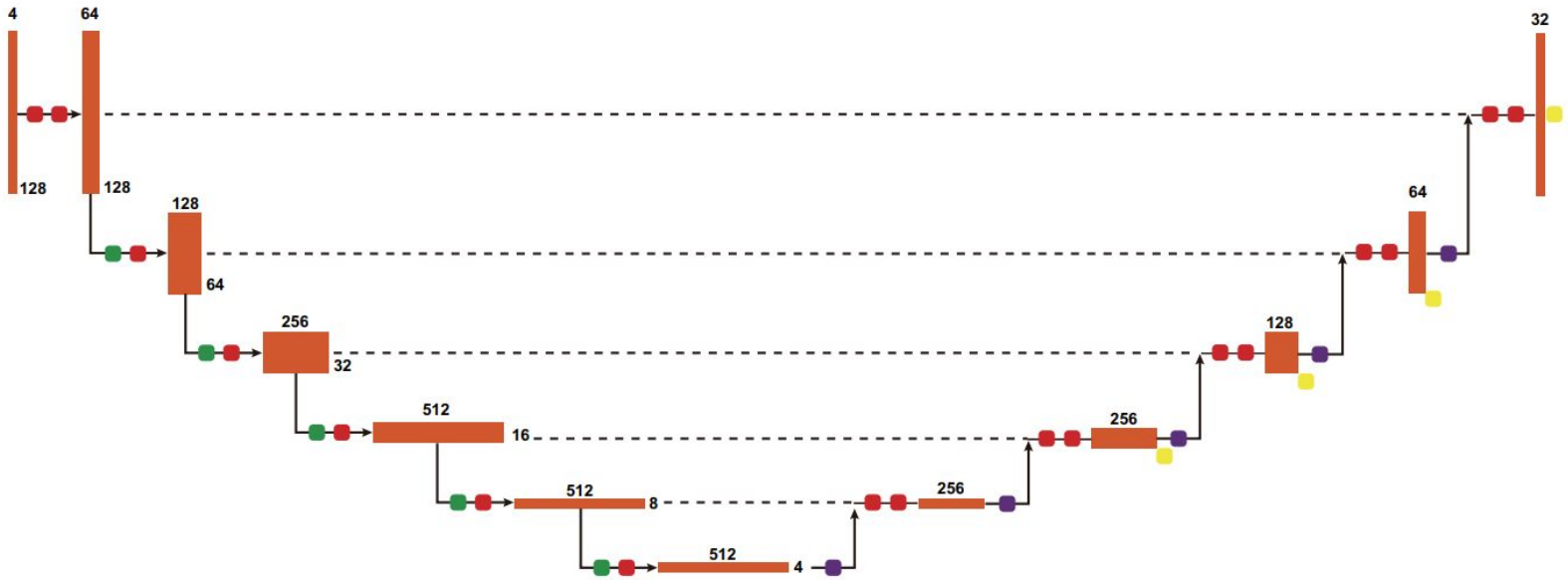
nnU-Net, region-based training

nnU-net trains on this



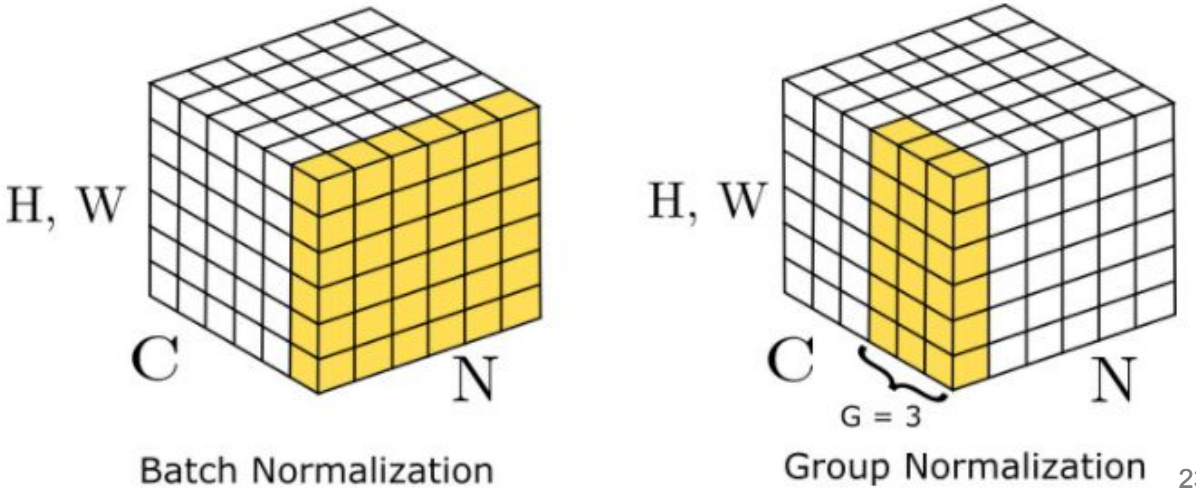
Proposed changes: Larger network

- Amount of training data = 4 x previous year
- Increasing the capacity of the network
- Doubling # of filters in encoder



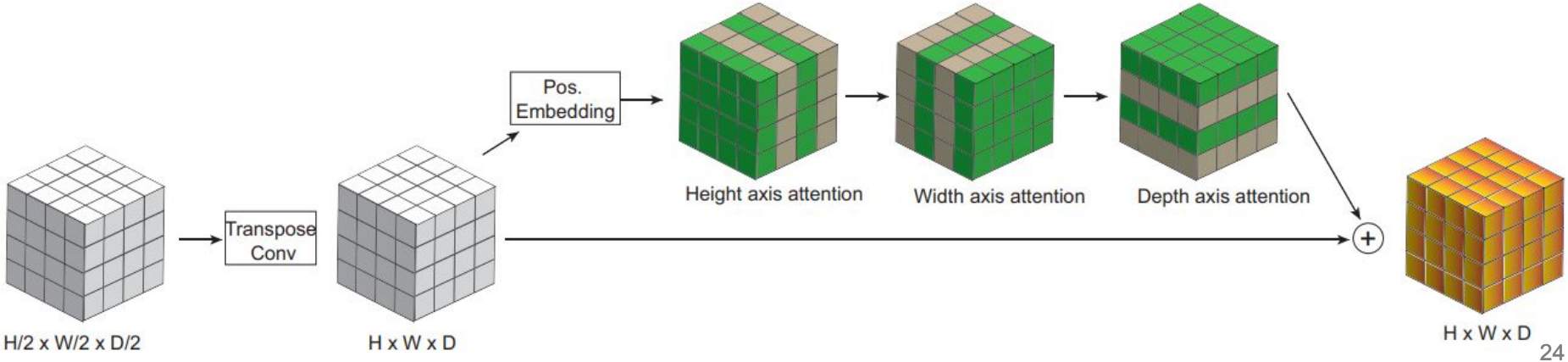
Proposed changes: GN instead of BN

- 3D conv networks demands lots of GPU memory
- This limit batch size
- GN shown to work better than BN for low batch size
- → Replace batch norm by group norm



Proposed changes: Axial attention decoder

- The challenge: Conv is only local
- Usually self-attention is used as a solution
- But self-attention is so costly for 3d images
- → Axial attention in decoder: one axis at a time



Methodology

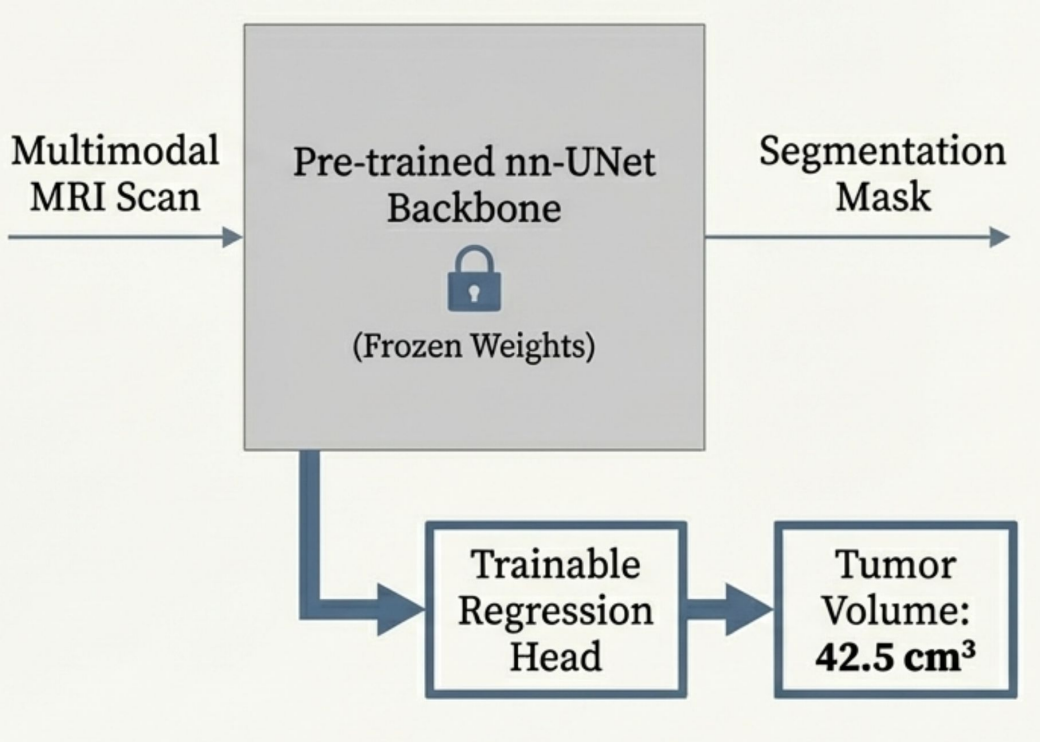
Extending nnU-Net for Brain Tumor Segmentation

Core Proposal:
We extend the winning architecture by adding a regression branch to directly predict numerical values (e.g., tumor volume) from MRI scans.

baseline:

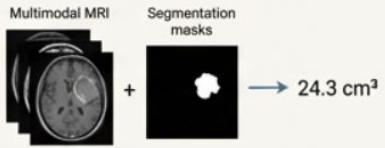
nnU-Net (3D U-Net self-configuring)

Key Advantage:
By freezing the pre-trained encoder/decoder, we reuse powerful feature extraction while significantly reducing training time and data requirements.



Project Methodology: A Step-by-Step Approach

1. Data Preparation



Construct a dataset where each multimodal MRI sample is paired with a ground-truth numerical label. This label (e.g., tumor volume) is calculated from the existing segmentation masks.

2. Model Training & Optimization

Train "only the newly added regression head on the prepared dataset. A suitable loss function, such as Mean Absolute Error (MAE) or Mean Squared Error (MSE), will be used to optimize for numerical prediction accuracy.

3. Evaluation & Interpretation

Rigorously evaluate the model's performance on a held-out test set using standard regression metrics. The numerical output can also be used to place patients into clinical categories (e.g., 'Stable', 'Under Control', 'Critical').

1. Data Preparation/Datasets and Preprocessing

- Used the BraTS 2021 multi-modal MRI dataset (T1, T1ce, T2, FLAIR) with manual tumor sub-region annotations.
- Applied standard preprocessing: 1×1×1 mm^3 resampling, skull-stripping, z-score normalization, and crop/pad to fit 3D network input.
- Converted raw BraTS data into the nnU-Net task format under nnUNet_raw_data.
- Organized data into imagesTr / imagesTs / labelsTr, saved as 3D NIfTI, with integer labels starting at 0 and metadata stored in dataset.json.

1. Data Preparation/Dataset Conversion for nnU-Net

BraTS → nnU-Net compatibility

- Converted the raw **BraTS** dataset into the **nnU-Net dataset format**.

Data format

- All images are stored as **3D NIfTI** files.
- Multi-modal scans have **consistent geometry across modalities**.

Label format

- Label maps use **consecutive integer IDs**, starting from **0** for background.

Metadata (dataset.json)

  - modality identifiers,
  - case lists,
  - dataset description structure aligned with the **Medical Segmentation Decathlon** format.

Note for 2D datasets

- 2D images can be converted to **pseudo-3D** by adding a **singleton dimension** and saving as **NIfTI**.

1. Data Preparation/Tumor Volume Extraction from Segmentation Masks

- Computed quantitative tumor volumes from ground-truth masks to create targets for a regression head.
- Extracted voxel spacing from NIfTI headers, formed a whole-tumor mask by merging labels 1, 2, and 4, then calculated volume and saved per-case results into a CSV.

Algorithm 2. Compute Whole Tumor Volume from Segmentation Mask

```
procedure WholeTumorVolumeMM3(seg_path)        ▷ Returns tumor volume in mm^3
    img ← load NIfTI image from seg_path
    seg ← img.get_fdata()
    voxel_spacing ← img.header.get_zooms()[: 3]    ▷ dx, dy, dz in mm
    voxel_volume ← product of voxel_spacing
    wt_mask ← voxels with labels {1, 2, 4}
    volume_mm3 ← sum of wt_mask × voxel_volume
    return volume_mm3
```

2. Model Training & Optimization/ Scalar Head for Tumor Size Estimation

- Compute tumor volume for each patient from ground-truth segmentation masks (to create targets for a tumor-size regression head).
- Extract voxel spacing from the NIfTI header and compute voxel volume.
- Build a whole-tumor mask by merging BraTS labels 1, 2, and 4.
- Calculate volume (mm³) as: # tumor voxels × voxel volume.
- Run this for all cases and save results to a CSV (patient ID + tumor volume).

Algorithm 1. NN-UNet Scalar Head for Tumor Size Estimation

```
procedure NNUNETSCALARHEAD(backbone, n_backbone_channels, hidden)
    Store backbone
    Initialize head ← Sequential(
        Flatten(),
        Linear(n_backbone_channels, hidden),
        ReLU(inplace=True),
        Linear(hidden, 1) )

    procedure FORWARD(x)
        feat ← backbone(x)
        if feat is tuple or list then
            feat ← feat[0]
        feat_summed ← Σ_{d,h,w} feat
        num_voxels ← x_D × x_H × x_W
        feat_avg ← feat_summed / num_voxels
        output ← head(feat_avg)
        return output

procedure _SET_REQUIRES_GRAD(module, requires_grad)
    for all p ∈ module.parameters() do
        p.requires_grad ← requires_grad
```

3. Evaluation & Interpretation

We used multiple metrics to assess volume prediction performance.

- **MAE (Mean Absolute Error):** Average absolute difference between **predicted** and **true** tumor volumes.
- **MAPE (Mean Absolute Percentage Error):** Relative error reported as a **percentage**.
- **R^2 (Coefficient of Determination):** Proportion of variance in the true volumes **explained by the model**.
- **CCC (Concordance Correlation Coefficient):** Measures the **agreement/consistency** between predicted and true volumes.
-

Result

Method 1 vs. Method 2: Volume Regression Results

	MAE (cm^3)	MAPE (%)	R^2	CCC
Method 1	26.3779	68.14	0.6340	0.7065
Method 2	8.8827	15.68	0.9601	0.9789

- **Method 1: AvgPool + Norm + MSE**
- **Method 2: SumPool + MLP + VoxelNorm + Log + Huber**

Method 1 vs. Method 2

Method 1 — AvgPool + Norm + MSE (Baseline)

- Backbone: frozen nnU-Net feature maps
- Feature aggregation: Average Pooling over spatial voxels → one global feature vector
- Head: simple regression head (lightweight)
- Normalization: basic feature/target normalization
- Loss: MSE
- Main limitation (why it underperforms):
  - AvgPool can dilute tumor-related signal, especially for small tumors (background dominates).
  - MSE is sensitive to outliers → a few large-volume cases can bias training.
  - Harder to handle skewed volume distribution (many small vs few large).

Method 2 — SumPool + MLP + VoxelNorm + Log + Huber (Improved)

- Backbone: frozen nnU-Net feature maps (same as Method 1)
- Feature aggregation: Sum Pooling over voxels → preserves “mass/extent” information
- Head: MLP (multi-layer) → captures non-linear mapping from features to volume
- VoxelNorm: normalization tied to voxel statistics / feature scale → more stable across cases
- Target transform: log(volume) → reduces scale imbalance & stabilizes learning
- Loss: Huber loss → robust to outliers (between MAE and MSE)

Why Method 2 better than Method 1

Why Method 2 Performs Better (Key Takeaways)

- SumPool > AvgPool: keeps volume-related signal instead of averaging it away
- MLP > simple head: learns richer, non-linear relationships
- Log target: balances the distribution of volumes → improves relative error (MAPE)
- Huber loss: less affected by extreme cases → improves training stability
- Overall: lower MAE/MAPE and higher R^2/CCC (better accuracy + agreement)

Error Histogram

error histogram

visualizes the distribution of prediction errors (e.g., predicted − actual)

X-axis (Error): magnitude and direction of errors (negative = underestimation, positive = overestimation).

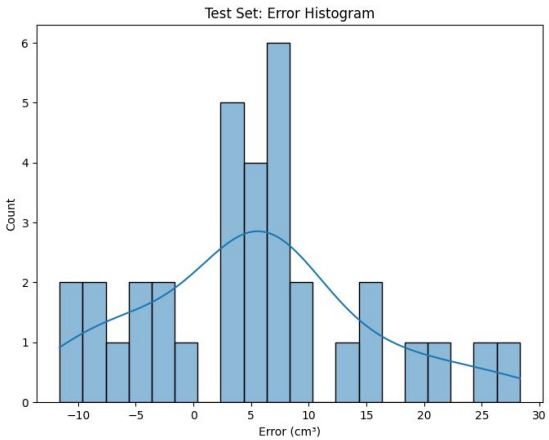
Y-axis (Count): frequency of errors within each bin.

Shape:

Centered near 0 → unbiased model.

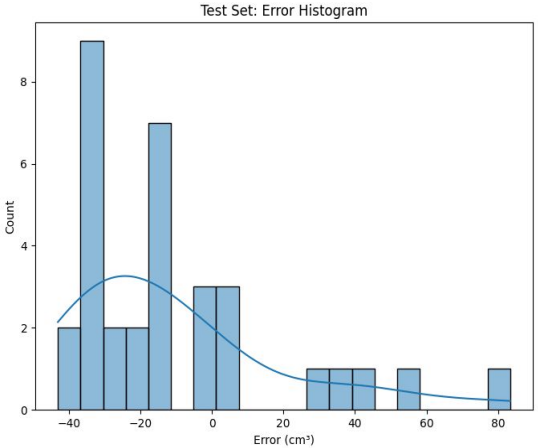
Spread (width) → variability/precision (narrow = more consistent).

Skew → systematic bias (more errors on one side).



First plot:

- **Bias :** centered slightly above 0 → mild overestimation bias.
- **Variance:**relatively narrow range (~ -10 to 30) → more consistent errors.
- **Skewness:** mildly right-skewed (some larger positive errors).
- **Outliers:**few moderate outliers.



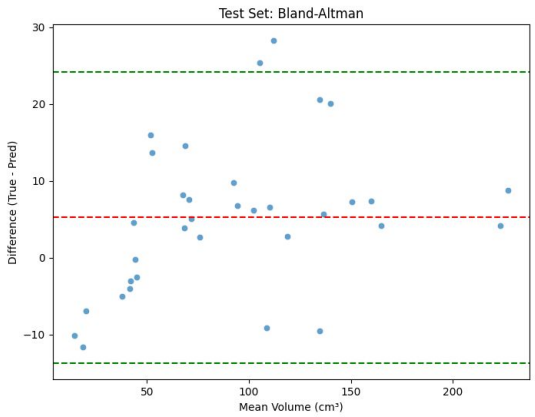
Second plot:

- **Bias :** centered below 0 → stronger underestimation bias.
- **Variance:**very wide range (~ -40 to 80) → high variability, less stable model.
- **Skewness:** strongly right-skewed (long positive tail, large overestimation outliers).
- **Outliers:** significant extreme values (especially large positive errors).

Bland-Altman

Bland–Altman

- Compares true and predicted values by plotting their difference vs. mean.
- The center line shows bias; the upper and lower lines (mean ± 1.96 SD) show limits of agreement.
- A good model has near-zero differences, narrow limits, and no trend.



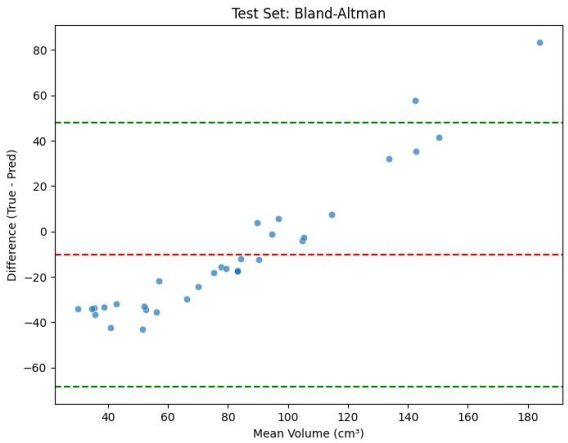
First plot:

- The bias is small and slightly positive.
- The limits of agreement are relatively tight.
- No strong trend is observed across the range.

Indicates fairly consistent and acceptable agreement.

Second plot:

- The bias is negative.
- The spread of differences is much larger.
- Differences increase as the mean increases (presence of proportional error).



Indicates higher variability and poorer agreement.

Train vs Validation HuberLoss

Huber Loss :

- Uses **L2 (MSE behavior)** for small errors and **L1 (MAE behavior)** for large errors
- Reduces the impact of outliers compared to MSE
- Smooth and stable for gradient-based optimization (controlled by threshold δ)

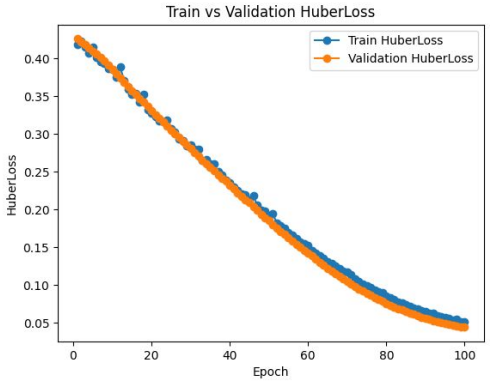
MSE (Mean Squared Error)

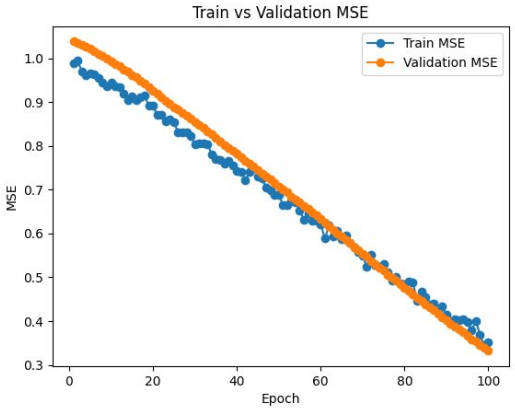
- Computes the mean of squared differences between predictions and true values
- Heavily penalizes large errors due to squaring
- Sensitive to outliers but encourages precise predictions

Difference :

- The **Fist plot (Huber Loss)** shows smoother and slightly more robust convergence because the loss reduces the impact of large errors.

- The **Second plot (MSE)** decreases more linearly but is more sensitive to outliers, which is why the curve shows slightly more variance.
  Overall, both train and validation curves are close , the model is learning well with no obvious overfitting.

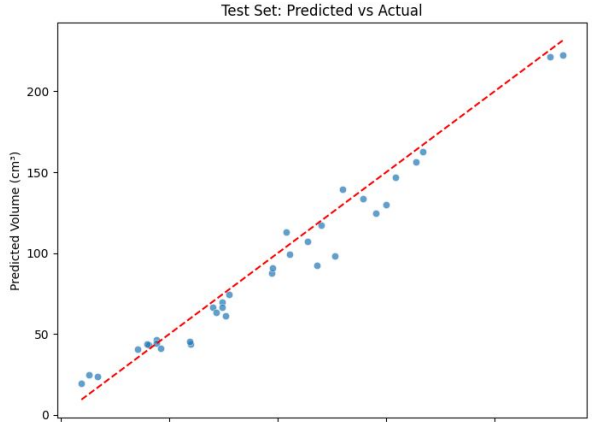


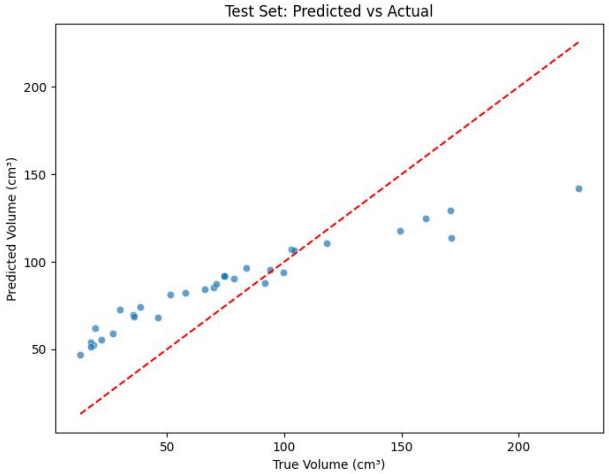


Predicted vs Actual

- **First plot:** Predictions lie close to the red diagonal (ideal line), meaning the model estimates volumes accurately with small errors. Performance looks good.

- **Second plot:** Points deviate more from the diagonal, especially at higher volumes , the model tends to **underestimate large values** and shows weaker accuracy.





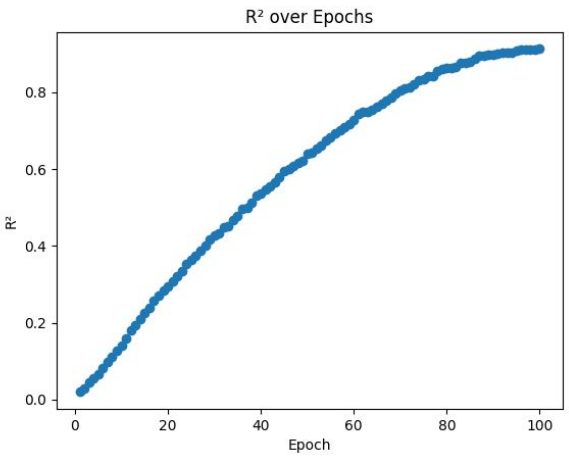
R^2

R^2 (R-squared)

R^2 (coefficient of determination) measures how well a model explains the variance of the target variable.

- Range: 0 to 1
- $R^2 = 0 \rightarrow$ model explains none of the variance
- $R^2 = 1 \rightarrow$ model explains all of the variance

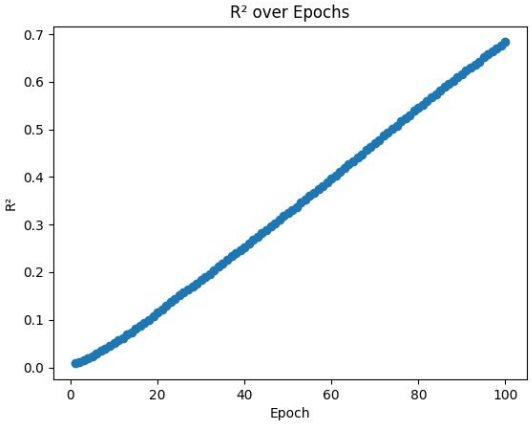
Higher R^2 means better fit to the data.



Difference :

- **First plot:** R^2 increases quickly at the beginning, then gradually slows down and plateaus near ~0.9. This suggests learning is strong early and then converges.

- **Second plot:** R^2 increases almost linearly and more slowly, ending around ~0.7. This suggests steady improvement but weaker overall performance compared to the first model.



CCC

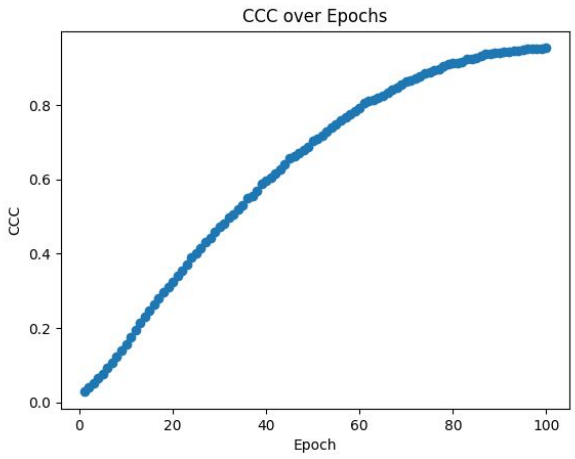
CCC (Concordance Correlation Coefficient)

CCC measures agreement between predicted and true values.

It considers both **correlation (precision)** and **mean difference (accuracy)**.

- Range: −1 to 1
- 1 → perfect agreement
- 0 → no agreement

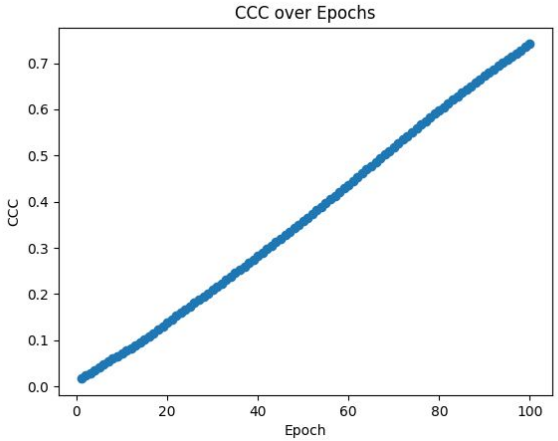
Higher CCC means better prediction quality.



Difference:

- **First plot:** CCC increases fast at the beginning and then slows, reaching about ~0.9. This shows strong learning and good final agreement.

- **Second plot:** CCC increases more linearly and ends around ~0.75. This indicates slower improvement and weaker overall performance compared to the first model.

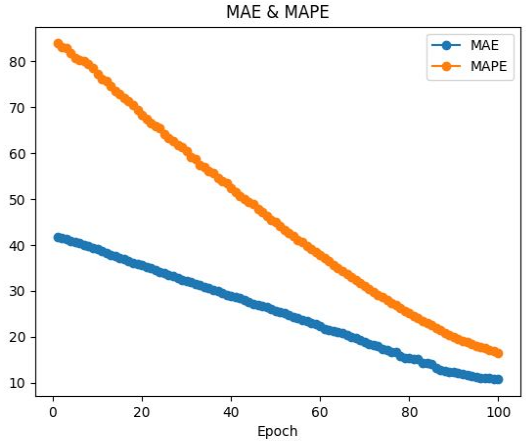


MAE and MAPE

MAE and MAPE

- **MAE (Mean Absolute Error):** Average absolute difference between predicted and true values. Lower MAE = better accuracy. Unit is the same as the target variable.

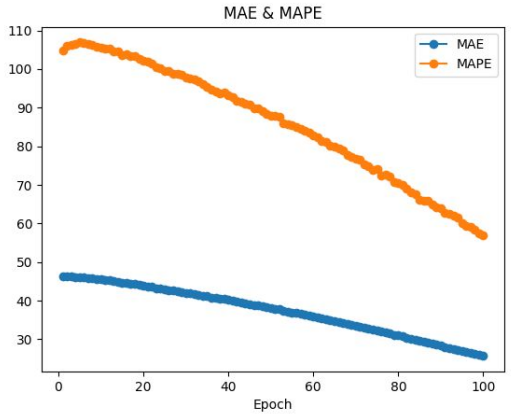
- **MAPE (Mean Absolute Percentage Error):** Average absolute percentage error between predicted and true values. Lower MAPE = better accuracy. Expressed in %.



Difference:

- **First plot:** MAE and MAPE decrease strongly and reach much lower final values. This indicates better model performance and faster improvement.

- **Second plot:** MAE and MAPE decrease more slowly and remain higher at the end. This shows weaker overall accuracy compared to the first model.



Conclusion & Future Work

- Extended **nnU-Net** for brain tumor segmentation with a lightweight **tumor volume prediction head**.
- Enabled **segmentation + volume estimation** in one framework with **minimal computational overhead**.
- Results suggest accurate segmentation can support **quantitative tumor burden assessment** without heavy architectures.
- Main limitation: **restricted hardware** (GPU/CPU), limiting tuning and long training → not fully optimized performance.

Future work:

- train on the **full dataset**, run **more epochs** and stronger tuning.
- Explore **multi-task learning** for segmentation + **sub-region volume** prediction.
- Study **longitudinal MRI** for tumor progression tracking.
- Test on **multi-center datasets** to evaluate generalization.
- Aim to integrate into **clinical decision-support systems**.

Thank you!

Q & A